SHEET INDEX

30 kW AC-DC (lower) - 1 of 3

AC-SENSING - 9 sections

AUX-POWER - 5 sections

CONTROL - 4 sections

INPUT-EMI - 4 sections

DC-LINK - 3 sections

VIENNA-PFC - 3 sections

rev D.4 - 28 sections - 341 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D1 LA0-LC0: 3x 0077908A7 KoolMu, N=39 (+/-1 lot trim), 3x(6x1mm) 18mm2 -> L0 169uH, >=75uH @82A pk (BIASED value governs ripple/trip calcs)

E68 EMI: CMC1 + 12x X2 4.7uF in 3 star stages + CMC2 + RC damper; D6 DM chokes deleted

Cr at TANK sections

DC-Modules 30 kW — AC-DC board (Vienna PFC)

DC-Modules 30 kW - board AC-DC (lower), sheet 1 of 3

2026-09-06 · rev D.4